

Josh Persi

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PROFESSIONAL SUMMARY

Data analyst with five years of experience and strong skills in the development of end-to-end ETL pipelines using enterprise task orchestrators. An academic background in biological sciences with the ability to think critically about complex interdependent systems. Significant experience using R for end-to-end analytical pipelines with a strong grasp of relational models, schema design, software engineering, and Azure. Adept at forming strong professional relationships with multidisciplinary teams to drive business goals.

EXPERIENCE

Science Specialist

Canadian Food Inspection Agency

Ottawa, ON

Oct. 2021 – Present

- Combed through existing ETL pipelines to identify and refactor inappropriate joins – resulting in 500k additional records being available
- Developed a novel processing pipeline to extract geographic coordinates for millions of text addresses – solving a years-long business problem by making these addresses usable for risk analytics
- Deployed Python scripts within an Azure environment for scheduled remote execution - ensuring laboratory data was always up to date
- Modernized the data science development life-cycle by implementing software engineering best practices, including version control with Git, automated testing, and reproducible virtual environments
- Spearheaded the adoption of Microsoft Fabric, leveraging a Lakehouse architecture similar to Databricks to unify disparate data streams into a single source of truth
- Leveraged compute and storage services from the Azure Cloud to analyze large global datasets
- Introduced both Dagster and Apache Airflow as solutions to orchestrate Agency ETL pipelines
- Implemented the first predictive capacity model for CFIA laboratories via an ARIMA time-series model to forecast annual work capacity
- Engineered an automated reporting pipeline by integrating SQL for retrieval of complex data and R for programmatic analysis, replacing a labor-intensive 3-day manual process
- Modelled lab technician performance using binomial logistic regression to analyze success rates, ensuring regulatory alignment with international seed testing compliance standards
- Determined statistically valid sample sizes for potato-wart resistance studies by performing power analyses to reach a target confidence level of 95%
- Introduced interactive, regularly updating Power BI reports as a solution to ad-hoc, manually generated data extracts – increasing data availability and saving at least 5 hours per week
- Collaborated with national stakeholders to identify gaps in a Canada-wide data-sharing system, translating complex requirements into custom-built analytical tools – formally recognized for Scientific Innovation for two consecutive years for significantly improving the system's utility and user engagement
- Successfully secured multi-year financial support and executive buy-in for ongoing analytics projects by demonstrating the direct impact of novel techniques on agency goals

EDUCATION

McGill University

Master of Science, Department of Biology

Montréal, QC

Jan. 2018 – Dec. 2020

University of Guelph

Bachelor of Science in Environmental Sciences, Department of Integrative Biology

Guelph, ON

Sep. 2013 – Apr. 2017